

Protactinium

Pa (Pa I). Classified transitions below 2 μm (levels and lines from the literature): 64 levels, 64 lines.

Pa⁺ ionisation limit 47500.00 cm⁻¹ = 5.88925 eV (210.526 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| er \| J' \rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (from the cited literature; NIST lists no excited levels) and measured lifetime (where one exists).

